

Awesome LUG Presentation

What does it take to build a distro?



How distros are put together, and the two ways to build one.

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OMAECHY

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debian



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01

Definition of a distro

What makes up a distro?

kernel

- + user space `coreutils, tools, libraries`
- + package manager
- + bootloader
- + desktop environment `optional`
- + installer `optional`



DEFINITION

Linux is only the kernel



As you may have heard, Linux is only the kernel, one part of a fully functioning operating system. When paired with a coreutils suite like GNU, it becomes a full OS, which while called Linux is, in fact, GNU/Linux.

The kernel schedules processes and drives hardware, and all other tools and applications plug into it, with the exception of drivers.

kernel

Processes, memory, filesystems, drivers.

user space

Shell, coreutils, libc, compilers, everything else.

02

Two methods of distro creation



From scratch

Putting everything together yourself

LFS

Compile the whole system from source, step by step. Obviously LFS users are the only *real* Linux users

Buildroot

Automated builds for embedded targets

Yocto

Professional framework for building embedded distros

This method is best for learning, embedded, and flexing on reddit



Derivative

Building off the work of someone more competent

Base Debian, Ubuntu, Arch btw, Fedora

Benefits Comes with packages, security updates, hardware support, and toxic users

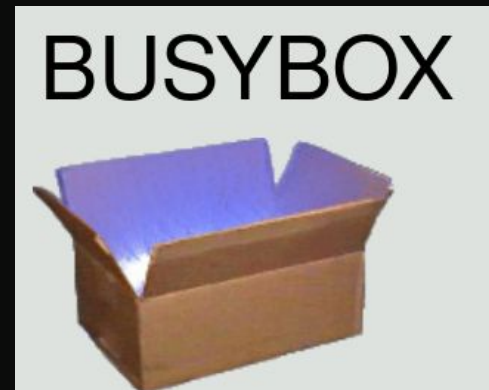
Tools CUBIC, archiso, debootstrap, live-build

This technique allows flexibility, letting us choose how much distro we're willing to make ourselves (in our case, very little)



From scratch vs. stealing

	From scratch	Derivative
Development time	Weeks	Hours (like 2 prompts)
Ongoing work required	Lots and lots	Your changes only
Knowledge gained	Enough to use Gentoo	Enough to larp on LinkedIn



GOAT

03

Core Components



BLOAT

Images to scale

CORE COMPONENTS

The Linux kernel

Clone the source, choose a config, compile. The config decides which drivers and features are in your OS at all.

A stock distro kernel supports everything. Yours only needs to support your target.



CORE COMPONENTS

Userspace

BusyBox

Hundreds of utilities in one small binary.

`embedded, rescue, containers,
based desktop setups`

GNU

Full coreutils, bash, glibc, toolchain.

`desktops, servers,
systemd shell setups`



💡 Tip: Don't break userspace or Linus Torvalds will yell at you on the mailing list

CORE COMPONENTS

The bootloader

GRUB	The default. Multi-boot, BIOS and UEFI, scriptable menus.
Syslinux	Small and simple. Good for a single-purpose image.
systemd-boot	UEFI only, almost no configuration.

```
Ubuntu 8.04, kernel 2.6.24-16-generic
Ubuntu 8.04, kernel 2.6.24-16-generic (recovery mode)
Ubuntu 8.04, memtest86+
Other operating systems:
Windows Vista/Longhorn (loader)
```

Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the
commands before booting, or 'c' for a command-line.

The highlighted entry will be booted automatically in 4 seconds.

CORE COMPONENTS

The Boot Chain

01 BIOS / UEFI firmware finds a boot device

02 Bootloader loads kernel + initramfs

03 Kernel takes over the hardware

04 initramfs mounts the real root filesystem

05 init systemd or OpenRC starts services

06 Login shell or desktop session

Package manager

Resolves dependencies, installs, upgrades, removes. Choosing a base distro means choosing one of these. Hamza will give you a lecture soon stay tuned

apt

Debian, Ubuntu

dnf

Fedora, RHEL

pacman

Arch (I use Arch btw)

Note these all use the old style of package management. You can also use binaries, Appimages, Snaps (hurray I love snaps), and Flatpaks (eww bloat)

apt - (A) (P)iece of (T)rash

dnf - (D)a (N)ifitiest (F)inder of the packages you want to install



Desktop environment or Tiling WM

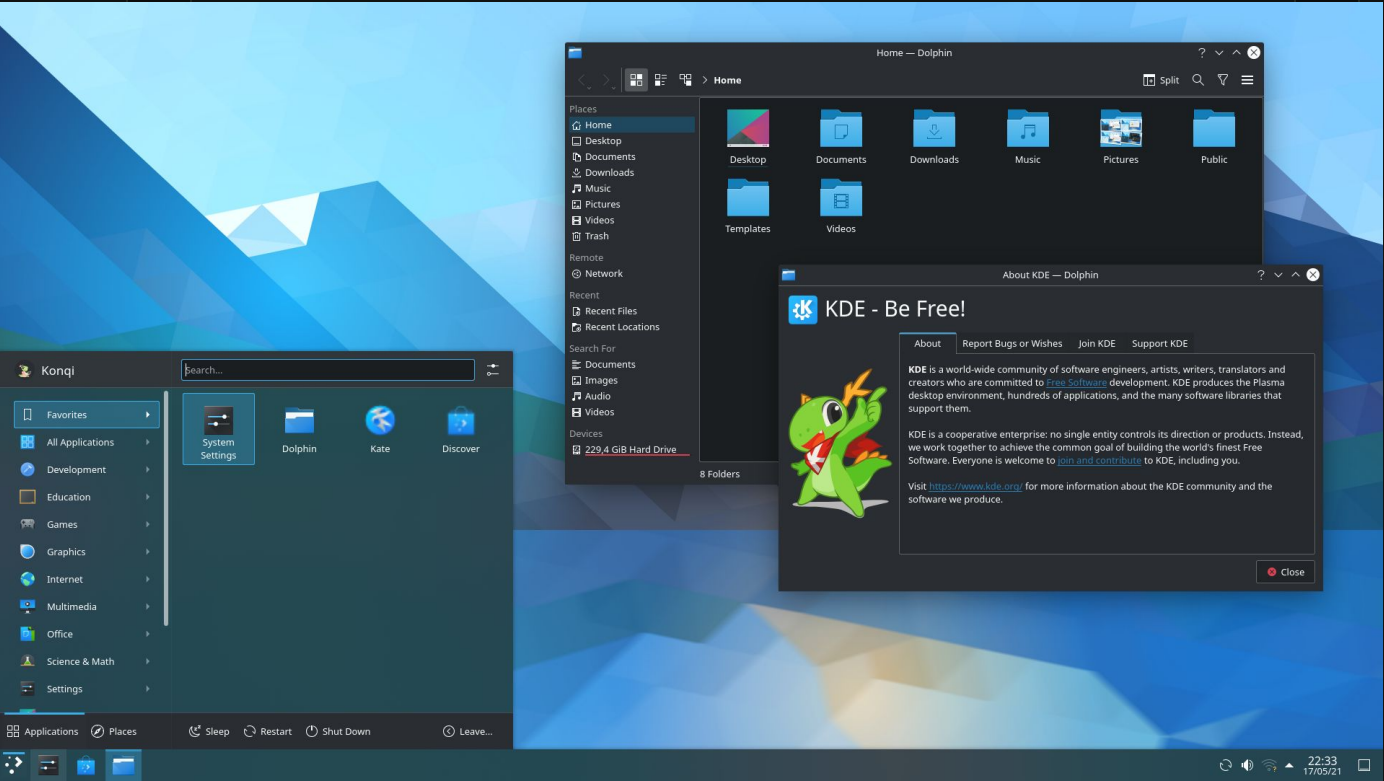
KDE. Well I *guess* you could use XFCE as well.



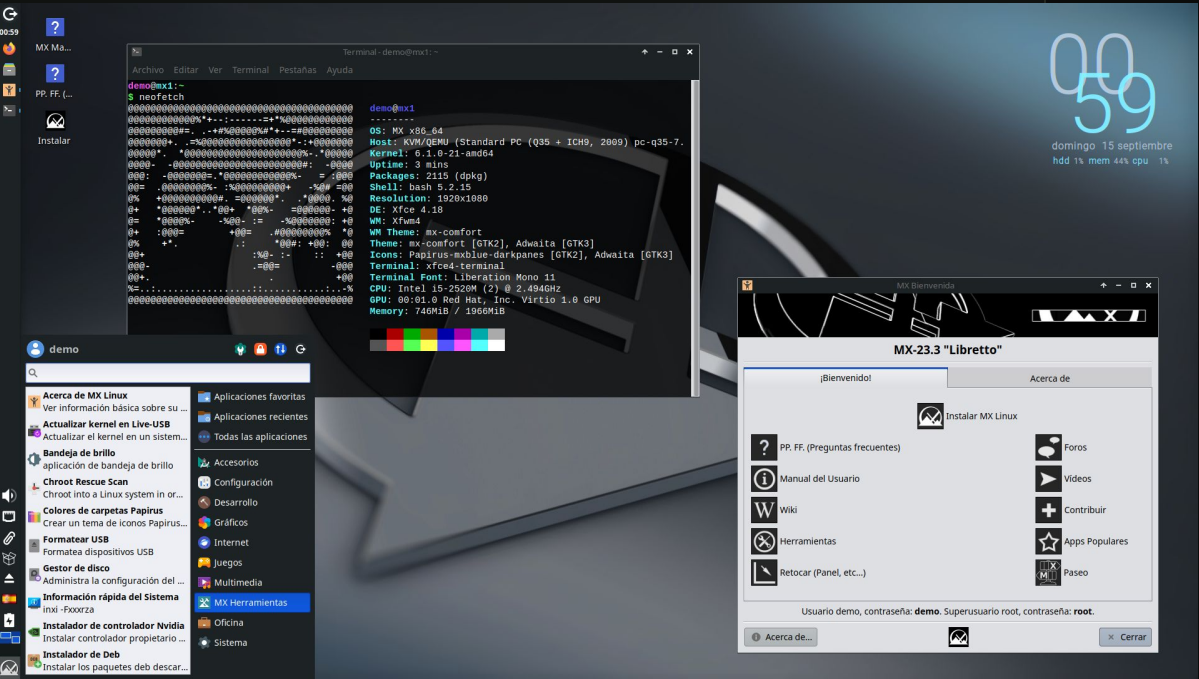
DWM (Superior)



GNOME screenshot



KDE screenshot



XFCE screenshot

The installer

Calamares

Modular, themeable, distro-agnostic

Anaconda

Fedora's installer, is very good

Scripts

Who does not love 1k+ lines of uncommented bash



04

Building from scratch

The steps required

- | | | | |
|----|---|----|---|
| 01 | Isolated build environment in a container | 02 | Install the toolchain and image tools |
| 03 | Configure and compile the kernel | 04 | Build BusyBox/GNU and pack an initramfs |
| 05 | Create and format a bootable disk image | 06 | Install and configure the bootloader |
| 07 | Boot it in QEMU | | |

The result?

A shell prompt. Ain't no way that was worth it.

To actually get a desktop system is a long and painful process. If you do want to go through that as a learning experience, check out the Linux From Scratch (LFS) project.



05

Building off an existing distro

Nine steps

- | | | | |
|----|---|----|--|
| 01 | Decide who it is for and what it does differently | 02 | Set up the build environment and tooling |
| 03 | Bootstrap a root filesystem from the base distro | 04 | Customize packages, desktop, users, defaults |
| 05 | Stand up a signed package repository | 06 | Add an installer |
| 07 | Build a hybrid BIOS + UEFI ISO | 08 | Test in a VM, then on real machines |
| 09 | Document it and publish the image | | |

06

HACKERMAN OS

Introducing: HACKERMAN OS

Are you ready for an entirely new unique incredible never-before-seen distro? Introducing - HACKERMAN OS



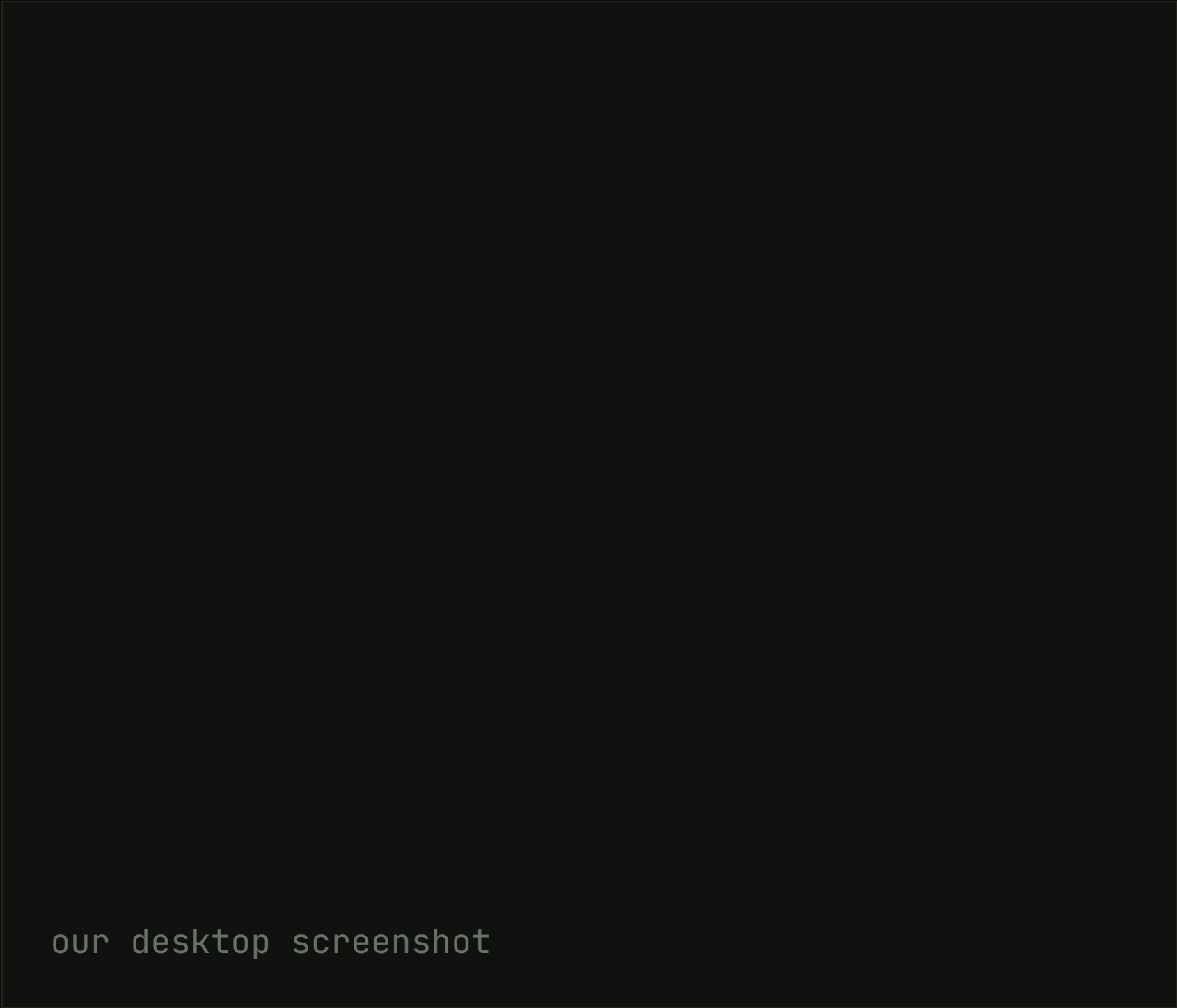
live boot / installer screenshot

OUR DISTRO

The tooling



build pipeline screenshot



our desktop screenshot

07

More info

Families of Distros

Independent	LFS · Gentoo · Alpine
Debian	Ubuntu · Mint · Kali · Pop!_OS
Arch	Manjaro · EndeavourOS
Fedora	RHEL · CentOS · Rocky

Other small insignificant things

Licensing

The kernel is GPLv2, because GPLv3 SUCKS or summin like the

Reproducibility

Core distros like Fedora and Ubuntu have extensive tooling to automatically build isos and packages

Security

Secure Boot, signed packages, disk encryption are painful to set up, so don't even bother

Community

Your distro will only have one user, so nobody's gonna want to switch unless you have like 12mil in funding. Still it's a fun project

Learning Path for the Club

First

Remaster Ubuntu with CUBIC. Read the FEDORA Wiki

Next

Work through LFS. Boot a kernel + BusyBox system

Then

Ship a derivative: signed repo, Calamares installer, custom ISO

Questions

Our presentation was flawless but perhaps you had issues understanding it

